Benchmark Items as Measurement Instruments: DIF Diagnostics for Response-Derived LLM Regimes on BBH

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Abstract

Large language model benchmarks are usually reported as aggregate leaderboard scores, but the underlying observations are item responses. I treat BBH as a measurement instrument: models are respondents, BBH questions are items, and the central question is whether items function similarly across model regimes after conditioning on overall performance. To study this, I build a harmonized Open LLM Leaderboard BBH analysis that compares (i) a simplified family-label replication baseline using Llama and Qwen labels, (ii) the original strict same-wrong MADNESS grouping, and (iii) score-residualized response clustering (SRRC), a refined response-derived grouping designed to reduce trivial score-based separation. The harmonized design uses two holdouts: response-derived groups are learned from 4,033 group-discovery items, while Mantel–Haenszel DIF is estimated on 1,727 disjoint DIF-test items; consequence analyses are then computed on held-out evaluation models. The family baseline recovers broad BBH DIF, with 49.3% Category-C items on the common DIF-test split. Strict MADNESS produces an even larger DIF surface but is diagnostically confounded with overall performance, leaving only 11 of 20 matching strata usable. SRRC yields a cleaner response-derived result: 43.2% Category-C items, all 20 strata usable, and held-out item-pool shifts of roughly $+13.6$ and -11.4 percentage points. The evidence supports measurement noninvariance as a useful diagnostic for benchmark validity in this BBH testbed, not a blanket claim that all leaderboard rankings are invalid.

1 Introduction

LLM evaluations are often summarized by a single benchmark score, but that score aggregates many item-level responses. This aggregation is useful for ranking models, yet it can hide whether different model populations reach the same score by succeeding on the same items. The relevant measurement question is therefore not only whether one group is more accurate on average, but whether the same items have the same conditional meaning across groups. In psychometrics, differential item functioning (DIF) formalizes this question: an item shows DIF when two groups with the same overall performance level have different probabilities of answering that item correctly (Holland and Thayer, 1988; Dorans and Holland, 1992; Millsap, 2012). This item-response framing also connects to recent work using psychometric tools to evaluate AI systems and LLM benchmarks, including IRT analyses of classifiers and benchmark instances, IRT-based efficient LLM evaluation, and statistical item diagnostics for benchmark validity (Martínez-Plumed et al., 2019; Polo et al., 2024; Truong et al., 2025; Zhou et al., 2026).

This project studies DIF in BBH (Suzgun et al., 2023), using item-level response data from the Open LLM Leaderboard setting (Fourrier et al., 2024). It was partly motivated by an unpublished family-label DIF analysis shared with me, in which Llama and Qwen are treated as known model

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groups. I use a simplified version of that family-label analysis only as a sanity-check baseline, not as the original contribution of this work. The main question here is whether comparable noninvariance can be detected when groups are learned from response behavior rather than supplied by model metadata.

The contribution is methodological rather than a new leaderboard claim. I use the family-label baseline to check that the local BBH response matrix and MH-DIF implementation recover a broad known-group signal. I then test two response-derived grouping strategies. The first is strict MADNESS, a same-wrong clustering approach inspired by prior work showing that LLMs have systematically correlated incorrect-answer patterns that can be used to measure model similarity and construct model taxonomies (Bradley, 2024). The second is score-residualized response clustering (SRRC), a refined response-derived grouping that residualizes broad score effects before clustering. The harmonized design evaluates the family baseline, strict MADNESS, and SRRC on the same held-out DIF-test items. The main conclusion is cautious: BBH item functioning can differ across response-derived regimes, but the interpretability of that signal depends strongly on how regimes are defined.

2 Data and harmonized design

BBH, or BIG-Bench Hard (Suzgun et al., 2023), is a collection of challenging BIG-Bench tasks designed to test reasoning, symbolic manipulation, and other capabilities that were difficult for earlier language models. In the Open LLM Leaderboard extraction used here, BBH is represented as binary item responses across 24 task domains, making it suitable for item-level DIF analysis. The final BBH response matrix used in this project contains 4,409 models and 5,760 items across 24 BBH domains. I use two independent split axes, summarized in Table 2. Models are split into 3,179 discovery models and 1,230 held-out evaluation models using seed 20260526. The split is proxy-cluster-aware, with 2,263 discovery clusters and 963 evaluation clusters, using model metadata and model-name patterns to reduce the chance that close derivatives are split across discovery and evaluation. Items are separately split into 4,033 group-discovery items and 1,727 DIF-test items, stratified by BBH domain.

Table 1: Harmonized split diagnostics. Response-derived groups are learned on group-discovery items; DIF is tested on DIF-test items.

Quantity	Count
Total BBH models	4409
Discovery models	3179
Evaluation models	1230
Total BBH items	5760
Group-discovery items	4033
DIF-test items	1727
Family baseline discovery: Llama / Qwen	1097 / 820
Family baseline evaluation: Llama / Qwen	355 / 335
Strict MADNESS decoded group-discovery items retained	3026

The two split axes serve different validity purposes. The item split controls circularity introduced by response-derived grouping: strict MADNESS and SRRC learn groups only from group-discovery items, and the DIF-test items are never used for grouping, tuning, centroid construction, or evaluation-model assignment. The model split controls circularity in the consequence analysis: DIF item pools are identified on discovery models, while item-pool consequences are evaluated on held-out models. This discovery/evaluation separation mirrors the family-label baseline: DIF item pools are defined on discovery models, and their consequences are assessed only on held-out evaluation models.

	Group-discovery items	DIF-test items
Discovery models	Learn response-derived group rules: strict MADNESS same-wrong groups and SRRC residualized centroids.	Estimate MH-DIF, assign ETS A/B/C categories, and define invariant / favoring item pools.
Evaluation models	Apply frozen group-assignment rules to held-out models without using DIF-test items.	Evaluate held-out item-pool consequences on the same DIF-test items.

Family baseline skips the group-learning cells because Llama/Qwen labels are external, but it uses the same DIF-test items and held-out evaluation logic.

Table 2: Harmonized design. The item split separates group discovery from DIF testing; the model split separates DIF item-pool identification from held-out consequence evaluation.

There is one important comparability caveat. The family baseline is restricted to the Llama/Qwen subset because family labels define the groups. This subset contains 1,097 Llama and 820 Qwen discovery models, plus 355 Llama and 335 Qwen evaluation models. The response-derived analyses use the full BBH model pool, because their groups are inferred from response behavior rather than architecture metadata. Therefore, the rows of Table 3 share the same DIF-test items and MH/ETS procedure, but they are not all estimated on the identical model population.

3 Methods

3.1 Mantel–Haenszel DIF and ETS categories

For each item j, let $Y_{ij} \in \{0, 1\}$ denote whether model i answers the item correctly, G_i denote group membership, and T_i denote a matching score summarizing overall performance. An item is invariant if

$$P(Y_{ij} = 1 \mid T_i = t, G_i = R) = P(Y_{ij} = 1 \mid T_i = t, G_i = F), \quad \text{for matched score levels } t,$$

where R is the reference group and F is the focal group. I use the Mantel–Haenszel common-odds-ratio estimator (Mantel and Haenszel, 1959; Holland and Thayer, 1988). Effects are reported on the ETS delta scale,

$$\Delta_{MH,j} = -2.35 \log \hat{\alpha}_{MH,j},$$

with positive values favoring the focal group. The focal group is Qwen for the family baseline, G1 for strict MADNESS, and EPn1 for SRRC.

I classify items into ETS-style categories using the same thresholds as the family-label DIF pipeline: Category A is negligible, defined by $|\Delta_{MH}| < 1$ or adjusted $p > 0.05$; Category C is large, defined by $|\Delta_{MH}| \geq 1.5$ and adjusted $p \leq 0.05$; Category B is the remaining moderate region (Dorans and Holland, 1992; Zieky, 2003). P-values are adjusted with Benjamini–Hochberg FDR control (Benjamini and Hochberg, 1995). All analyses use 20 score bins. The anchor20 pass follows a two-stage procedure: run an initial total-score MH pass, select the lowest-$|\Delta_{MH}|$ 20% of DIF-test items as anchors, construct anchor-only matching strata, and recompute MH-DIF with BH-FDR.

The MH/ETS implementation is intentionally simple. It uses the MH effect estimator, ETS delta scale, A/B/C categorization, BH-FDR, and anchor20 matching logic, but it does not implement cluster-robust Phillips–Holland/Rao–Scott variance corrections (Phillips and Holland, 1987; Rao and Scott, 1992). This matters because Open LLM Leaderboard models include many fine-tunes and descendants from common base models. I therefore interpret significance-based DIF categories as descriptive diagnostics rather than as fully cluster-robust inferential claims.

3.2 Group definitions

Family-label baseline. The family-label baseline uses externally supplied Llama/Qwen labels and has no group-learning step. It is included first as a replication-style sanity check: it tests whether the local BBH response matrix and MH-DIF implementation recover a broad family-level DIF signal for an externally defined Llama/Qwen contrast.

Strict MADNESS. Strict MADNESS operationalizes the same-wrong intuition from Bradley (2024) as a response-derived grouping procedure for this project. It clusters models by same-wrong behavior: two models are similar when they repeatedly choose the same incorrect option. In the harmonized run, group learning uses only discovery models and group-discovery items with decoded option data. Of the 4,033 group-discovery items, 3,026 are retained after decoded-sidecar filtering. Items must have at least two answer options and at least two models with valid decoded prediction/response cells. Models must have at least $\max(50, 0.8 \times 3026) = 2420$ valid decoded group-discovery cells. Ten low-coverage models are left unassigned. Same-wrong z-similarity was symmetrized and converted into a dissimilarity,

$$D_{ij} = z_{\max} - \{z_{ij} + z_{ji}\}/2,$$

where $z_{\max}$ is the maximum observed same-wrong similarity. Ward-style hierarchical clustering is then applied to this dissimilarity and cut into two groups. Because this dissimilarity is not guaranteed to be Euclidean, strict MADNESS is treated as a heuristic diagnostic rather than a fully principled clustering model.

SRRC. SRRC is a refinement introduced after strict MADNESS diagnostics indicated strong ability confounding. It still learns groups from responses, but first constructs residualized domain-response features. The residualization removes a cubic function of overall score, after which k-means is fit on discovery models using group-discovery items only. The selected solution is $k = 2$: it has nearly balanced discovery groups, low ability predictability ($R^2 = 0.0027$), and high bootstrap stability (mean ARI $= 0.998$; minimum ARI $= 0.992$) among the candidate solutions recorded in the output. Evaluation models are assigned by applying the frozen residualization/standardization transform to group-discovery items and assigning each model to the nearest frozen centroid. No DIF-test item is used in this assignment.

3.3 Held-out consequence analysis

For each analysis, DIF effects and ETS categories are estimated on discovery models using the common 1,727 DIF-test items. These classifications define item pools: all DIF-test items, invariant A items, B/C items favoring the focal group, and B/C items favoring the reference group. Consequences are evaluated only on held-out evaluation models. The gap sign is Qwen–Llama for the family baseline, G1–G0 for strict MADNESS, and EPn1–EPn2 for SRRC. The reported draw summaries use 500 random 50-item draws from each pool.

4 Results

Table 3: Harmonized MH-DIF results on the common 1,727-item DIF-test split.

Analysis	Items	A	B	C	Median $\lvert\Delta_{MH}\rvert_C$	Usable strata
Family baseline	1727	617 (35.7%)	258 (14.9%)	852 (49.3%)	2.47	20/20
Strict MADNESS	1727	461 (26.7%)	185 (10.7%)	1081 (62.6%)	3.28	11/20
SRRC	1727	674 (39.0%)	307 (17.8%)	746 (43.2%)	2.27	20/20

4.1 Family-label baseline recovers a broad known-group DIF signal

The family baseline is the least circular comparison because its groups are external labels rather than response-derived clusters. On the common 1,727-item DIF-test set, it finds 617 A items (35.7%), 258 B items (14.9%), and 852 C items (49.3%), with median $|\Delta_{MH}| = 2.47$ among Category-C items. This result serves as a local sanity check: the Llama/Qwen contrast produces a broad family-label DIF surface under the same BBH processing and MH/ETS machinery used for the response-derived analyses. Because this baseline is simplified and does not include cluster-robust inference, I use it to verify pipeline behavior rather than as an exact reproduction of the unpublished analysis.

4.2 Strict MADNESS detects large DIF but has poor score-stratum overlap

Strict MADNESS produces the largest raw DIF surface: 461 A items (26.7%), 185 B items (10.7%), and 1,081 C items (62.6%), with median $|\Delta_{MH}| = 3.28$ among Category-C items. The high

Category-C rate should not be interpreted as cleaner evidence of measurement noninvariance. The matching diagnostics show that the same-wrong groups are highly aligned with overall score. Only 11 of 20 anchor strata are usable. Low-score bins are almost entirely G0, while high-score bins are almost entirely G1.

Table 4: Strict MADNESS anchor20 stratum diagnostics. Only 11 of 20 bins are usable because the low and high score tails are dominated by one group.

Score bin	G0	G1	Usable
0	166	0	No
1	163	0	No
2	167	0	No
3	147	0	No
4	155	1	No
5	147	7	No
6	120	41	Yes
7	133	40	Yes
8	120	60	Yes
9	97	60	Yes
10	73	69	Yes
11	54	102	Yes
12	67	114	Yes
13	73	64	Yes
14	91	88	Yes
15	33	105	Yes
16	11	146	Yes
17	0	176	No
18	1	133	No
19	6	144	No

This explains why strict MADNESS remains secondary in the final interpretation. It did detect a strong difference between groups, but because the groups behave like an ability split, the resulting DIF flags mix item-level noninvariance with broad performance separation. The analysis is therefore useful as a diagnostic first attempt and as the motivation for SRRC, not as the main response-derived claim.

4.3 SRRC gives the cleanest response-derived result

Table 5: Response-derived group sizes. Strict MADNESS leaves five models unassigned in each model split because of decoded-coverage filtering; SRRC assigns all models by frozen centroids.

Analysis split	Group 1	Group 2	Unassigned
Strict MADNESS discovery	G0: 1824	G1: 1350	5
Strict MADNESS evaluation	G0: 310	G1: 915	5
SRRC discovery	EPn1: 1621	EPn2: 1558	0
SRRC evaluation	EPn1: 645	EPn2: 585	0

SRRC yields a more interpretable response-derived grouping. As shown in Table 5, its discovery and evaluation groups are both relatively balanced, and all evaluation models are assigned by the frozen centroid rule. On the common DIF-test set, SRRC yields 674 A items (39.0%), 307 B items (17.8%), and 746 C items (43.2%), with median $|\Delta_{MH}| = 2.27$ among Category-C items. All 20 score strata are usable.

This result is smaller than strict MADNESS by Category-C prevalence, but more interpretable as a DIF analysis because the score-matching strata retain overlap between groups. SRRC uses the same item-holdout protection, assigns held-out models using a frozen discovery-only rule, retains all matching strata, and was selected specifically to reduce ability predictability. I therefore treat SRRC as the main response-derived analysis.

4.4 Selected item pools shift held-out group gaps

The held-out consequence analysis further distinguishes strict MADNESS from SRRC. For the family baseline, the full-pool Qwen–Llama gap on the DIF-test items is +2.55 percentage points. B/C pools favoring Qwen move the held-out gap to +18.70 points, while B/C pools favoring Llama move it to −12.33 points. For SRRC, the full-pool EPn1–EPn2 gap is only +1.33 points, while B/C favoring pools move it to +13.63 and −11.38 points. Invariant A pools are much closer to the full-pool gap: +1.12 for the family baseline and +0.89 for SRRC.

Strict MADNESS also shows selected-pool movement, but the interpretation is less clean. The full-pool G1–G0 gap is already +18.29 points. Even G0-favoring B/C items leave a positive +7.16 point gap. This is consistent with the stratum diagnostics: strict MADNESS groups are separated so strongly by overall performance that item-pool shifts do not isolate measurement noninvariance as cleanly as SRRC.

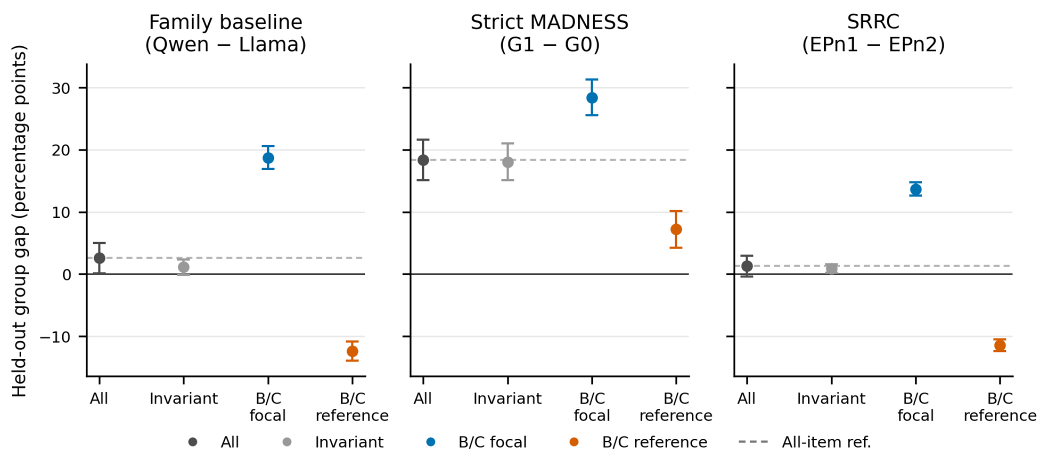

Figure 1: Held-out consequence of DIF-defined item pools. Points show mean held-out group gaps across 500 random 50-item draws; error bars show one standard deviation. The family baseline uses Qwen–Llama, strict MADNESS uses G1–G0, and SRRC uses EPn1–EPn2. All methods use the same 1,727 DIF-test items. Strict MADNESS shows broad performance separation, whereas SRRC shows cleaner opposite-direction shifts.

Figure 1 summarizes the consequence analysis: family and SRRC DIF-selected pools move the held-out gap in opposite directions, whereas strict MADNESS retains a large positive gap even for reference-favoring pools.

5 Discussion

The harmonized analysis clarifies why the grouping method matters. The family-label baseline verifies that the local BBH data pipeline and MH/ETS implementation recover a broad known-group DIF signal. Strict MADNESS shows that naive same-wrong clustering can find large group differences, but those differences are difficult to interpret when score-stratum overlap is poor. SRRC gives a cleaner response-derived test: its residualized features preserve all matching strata, its frozen centroid rule assigns held-out models without using DIF-test items, and its all-item group gap is much smaller.

The distinction between strict MADNESS and SRRC is the main lesson of the project. In this report, a “regime” should therefore be read as an operational grouping induced by a specified response-pattern procedure, not as a causal claim about training lineage or model architecture. Response-derived grouping is not automatically a valid measurement-regime definition. If the grouping procedure mostly separates high-performing from low-performing models, then many items will look noninvariant even after score matching, because there may be little within-stratum overlap. A useful response-derived regime should be stable, assign held-out models without using test items, and leave enough overlap across matching strata for DIF to be interpretable. SRRC better satisfies these criteria in the present run.

The consequence analysis should also be interpreted carefully. The selected B/C item pools produce sizable held-out gap shifts, especially for the family baseline and SRRC. That shows that the DIF surface is not merely a list of statistical flags: item composition can change group-level comparisons on held-out models. However, this is not a full rank-stability analysis over all models, and it does not prove that the global BBH leaderboard is invalid. It shows that model-regime comparisons can depend on which items are emphasized. Because I did not run difficulty-matched resampling baselines, these shifts should be interpreted as consequences of DIF-defined item pools rather than as effects fully separated from possible pool-level difficulty differences.

Several limitations remain. The analysis is BBH-only, so it does not establish whether the same response-derived regimes generalize to GPQA, MMLU-Pro, MATH, IFEval, MuSR, or other benchmarks. The family baseline is a simplified replication; it lacks the cluster-robust variance correction used in prior work, so the reported p-values may be overconfident in the presence of base-model descendants. As a sensitivity check, replacing the current simplified Wald inference with a CMH-style inference produced nearly identical A/B/C summaries: Category-C counts were unchanged for the family baseline and SRRC and differed by only two items for strict MADNESS.

The response-derived methods also have design-specific limitations. Strict MADNESS depends on decoded answer-option availability and drops or leaves unassigned low-coverage models. SRRC was introduced after diagnosing strict MADNESS, so it is best understood as an exploratory refinement rather than a fully pre-registered confirmatory method. Finally, the current report evaluates item-pool consequences but does not include Spearman/Kendall rank stability, top-k overlap, or model-based predictive-fit comparisons.

6 Conclusion

This project studies whether DIF diagnostics can be applied not only to externally defined model families, but also to response-derived model regimes. The final design separates group learning from DIF testing with an item split and separates DIF item-pool identification from consequence evaluation with a model split. Under this design, BBH item functioning differs not only across external Llama/Qwen labels but also across a refined response-derived regime definition. Future work should test whether SRRC-style regimes transfer beyond BBH, compare them against family, provider, size, and accuracy-bin baselines, and evaluate benchmark-level consequences through rank stability and predictive-fit analyses. The safest conclusion is that measurement noninvariance is a useful diagnostic lens for LLM benchmark validity: it can reveal when aggregate scores summarize different item-level patterns across model regimes. The result is not that leaderboards should be discarded, but that item-level invariance checks should accompany aggregate benchmark comparisons when those comparisons are used to support claims about model populations.

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